

PubH7485 Homework5 Answerkey

2024-11-01

Question 1 (10 points)

Table 1: Summary Statistics Stratified by the use of corticosteroids

	level	no	yes	p	test	SMD
n		495	877			
active (%)	placebo	214 (43.2)	370 (42.2)	0.750		0.021
	active	281 (56.8)	507 (57.8)			
gender (%)	male	275 (55.6)	514 (58.6)	0.297		0.062
	female	220 (44.4)	363 (41.4)			
age (mean (SD))		56.92 (15.48)	55.80 (14.71)	0.182		0.075
Asthma (%)	no	444 (89.7)	787 (89.7)	1.000		0.001
	yes	51 (10.3)	90 (10.3)			
COPD (%)	no	470 (94.9)	816 (93.0)	0.200		0.080
	yes	25 (5.1)	61 (7.0)			
Diabetes requiring medication (%)	no	380 (76.8)	643 (73.3)	0.179		0.080
	yes	115 (23.2)	234 (26.7)			
CHF (%)	no	475 (96.0)	843 (96.1)	0.996		0.008
	yes	20 (4.0)	34 (3.9)			
Hypertension requiring medication (%)	no	282 (57.0)	482 (55.0)	0.507		0.040
	yes	213 (43.0)	395 (45.0)			
Malignancy (active or receiving treatment) (%)	no	471 (95.2)	847 (96.6)	0.245		0.072
	yes	24 (4.8)	30 (3.4)			
Renal impairment (%)	no	451 (91.1)	796 (90.8)	0.907		0.012
	yes	44 (8.9)	81 (9.2)			
Immunosuppressed (%)	no	446 (90.1)	812 (92.6)	0.133		0.089
	yes	49 (9.9)	65 (7.4)			
bmi (mean (SD))		30.63 (7.48)	32.74 (8.75)	<0.001		0.259
New Score (mean (SD))		3.40 (2.34)	4.17 (2.05)	<0.001		0.353
Creatinine (mg/dL) (mean (SD))		1.14 (1.23)	1.16 (1.67)	0.842		0.012
C-reactive protein (mean (SD))		81.95 (71.30)	92.94 (68.75)	0.005		0.157
Lymphocytes (10 ⁹ /L) (mean (SD))		1.15 (1.65)	1.01 (2.80)	0.308		0.061
Pulm ordinal (%)	no O2	254 (51.3)	152 (17.3)	<0.001		0.790
	low O2	163 (32.9)	412 (47.0)			
	high O2	78 (15.8)	313 (35.7)			
Log 10 of Viral load copies/mL (mean (SD))		4.32 (1.83)	4.21 (1.76)	0.298		0.058

Question 2 (5 points)

The table below presented the unadjusted mean difference in hospitalized free days and 95% CI. The difference is significant since the 95% CI does not contain 0.

Table 2: Unadjusted ATE for Hospital-Free Days

Type	Estimate	SE	CI_Lower	CI_Upper
ATE	-1.112	0.416	-1.927	-0.296

Question3 (10 points)

The ATE and ATT estimates using regression model are presented below. The differences are not significant since both confidence intervals contain 0.

Table 3: ATE and ATT Estimates from Regression Model

Type	Estimate	SE	CI_Lower	CI_Upper
ATE	0.184	0.364	-0.529	0.897
ATT	0.148	0.362	-0.560	0.857

Note that I was using the regression model from stepwise model selection; the results may be slightly different when using other models.

Question 4 (35 points total; 5 points for a, 10 points for b, c, d)

a)

Table 4: Summary Statistics Stratified by the use of corticosteroids using IPW

	level	no	yes	SMD
n		1376.6	1368.9	
active (%)	placebo	622.4 (45.2)	565.1 (41.3)	0.079
	active	754.2 (54.8)	803.7 (58.7)	
gender (%)	male	794.3 (57.7)	794.3 (58.0)	0.006
	female	582.3 (42.3)	574.6 (42.0)	
age (mean (SD))		55.69 (15.20)	56.01 (15.03)	0.021
Asthma (%)	no	1227.6 (89.2)	1219.7 (89.1)	0.002
	yes	148.9 (10.8)	149.2 (10.9)	
COPD (%)	no	1292.7 (93.9)	1283.9 (93.8)	0.005
	yes	83.9 (6.1)	85.0 (6.2)	
Diabetes requiring medication (%)	no	1058.2 (76.9)	1011.7 (73.9)	0.069
	yes	318.3 (23.1)	357.2 (26.1)	
CHF (%)	no	1318.3 (95.8)	1316.8 (96.2)	0.022
	yes	58.3 (4.2)	52.1 (3.8)	
Hypertension requiring medication (%)	no	790.2 (57.4)	760.7 (55.6)	0.037
	yes	586.4 (42.6)	608.2 (44.4)	
Malignancy (active or receiving treatment) (%)	no	1329.2 (96.6)	1320.9 (96.5)	0.004

	level	no	yes	SMD
Renal impairment (%)	yes	47.3 (3.4)	48.0 (3.5)	0.095
	no	1278.7 (92.9)	1235.6 (90.3)	
Immunosuppressed (%)	yes	97.9 (7.1)	133.3 (9.7)	0.008
	no	1266.8 (92.0)	1256.9 (91.8)	
bmi (mean (SD))	yes	109.8 (8.0)	112.0 (8.2)	0.017
	no	32.16 (8.40)	32.02 (8.49)	
New Score (mean (SD))		3.88 (2.27)	3.90 (2.11)	0.010
Creatinine (mg/dL) (mean (SD))		1.08 (1.04)	1.14 (1.50)	0.044
C-reactive protein (mean (SD))		90.34 (74.36)	89.46 (67.24)	0.012
Lymphocytes (10 ⁹ /L) (mean (SD))		1.10 (1.28)	1.05 (3.21)	0.018
Pulm ordinal (%)	no O2	404.1 (29.4)	401.5 (29.3)	0.006
	low O2	583.2 (42.4)	576.7 (42.1)	
	high O2	389.3 (28.3)	390.7 (28.5)	
Log 10 of Viral load copies/mL (mean (SD))		4.19 (1.78)	4.22 (1.77)	0.018

b) The ATE and ATT estimates using PSS are presented below. The differences are not significant since both confidence intervals contain 0.

Table 5: ATE and ATT Estimates using PSS

Type	Estimate	SE	CI_Lower	CI_Upper
ATE	-0.1359536	0.4425787	-1.003392	0.7314848
ATT	-0.2042140	0.4935557	-1.171565	0.7631374

c) The ATE and ATT estimates using IPW are presented below. The differences are not significant since both confidence intervals contain 0.

Table 6: ATE and ATT Estimates using IPW

Type	Estimate	SE	CI_Lower	CI_Upper
ATE	-0.0825704	0.5861862	-1.231474	1.0663334
ATT	-0.2667100	0.4543694	-1.157258	0.6238377

d) i.

Table 7: Summary Statistics Stratified by the use of corticosteroids using 1:1 Matching

	level	no	yes	SMD
n		877	877	
active (%)	placebo	409 (46.6)	370 (42.2)	0.090
	active	468 (53.4)	507 (57.8)	
gender (%)	male	549 (62.6)	514 (58.6)	0.082
	female	328 (37.4)	363 (41.4)	
age (mean (SD))		55.56 (14.39)	55.80 (14.71)	0.017
Asthma (%)	no	781 (89.1)	787 (89.7)	0.022

	level	no	yes	SMD
COPD (%)	yes	96 (10.9)	90 (10.3)	
	no	805 (91.8)	816 (93.0)	0.047
Diabetes requiring medication (%)	yes	72 (8.2)	61 (7.0)	
	no	630 (71.8)	643 (73.3)	0.033
CHF (%)	yes	247 (28.2)	234 (26.7)	
	no	841 (95.9)	843 (96.1)	0.012
Hypertension requiring medication (%)	yes	36 (4.1)	34 (3.9)	
	no	480 (54.7)	482 (55.0)	0.005
Malignancy (active or receiving treatment) (%)	yes	397 (45.3)	395 (45.0)	
	no	852 (97.1)	847 (96.6)	0.033
Renal impairment (%)	yes	25 (2.9)	30 (3.4)	
	no	821 (93.6)	796 (90.8)	0.106
Immunosuppressed (%)	yes	56 (6.4)	81 (9.2)	
	no	825 (94.1)	812 (92.6)	0.059
bmi (mean (SD))		52 (5.9)	65 (7.4)	
		32.52 (8.21)	32.74 (8.75)	0.025
New Score (mean (SD))		4.06 (2.17)	4.17 (2.05)	0.053
Creatinine (mg/dL) (mean (SD))		1.09 (1.14)	1.16 (1.67)	0.050
C-reactive protein (mean (SD))		93.45 (72.34)	92.94 (68.75)	0.007
Lymphocytes (10 ⁹ /L) (mean (SD))		1.04 (0.67)	1.01 (2.80)	0.014
Pulm ordinal (%)	no O2	150 (17.1)	152 (17.3)	0.081
	low O2	445 (50.7)	412 (47.0)	
	high O2	282 (32.2)	313 (35.7)	
Log 10 of Viral load copies/mL (mean (SD))		4.17 (1.79)	4.21 (1.76)	0.023

- ii. The ATT estimate using 1:1 matching are presented below. The difference is not significant since the confidence interval contains 0.

Table 8: ATT Estimate using 1:1 Matching

Type	Estimate	SE	CI_Lower	CI_Upper
ATT	-0.197	0.928	-2.017	1.622

Question 5 (10 points)

The analyses above indicate that the adjusted ATE and ATT estimates using various causal methods were not significant, while the unadjusted ATE was significant. This highlights the importance of carefully selecting causal methods in non-randomized studies (and even in randomized studies when the factor of interest is not the primary treatment).

Appendix - R code

Question 1

```
# load and impute dataset
tico_7485 <- read.csv(file.choose(), header=T)
```

```

tico_7485$active <- factor(tico_7485$active, levels = c(0,1), labels = c('placebo', 'active'))
tico_7485$gender <- factor(tico_7485$gender, levels = c(1,2), labels = c('male', 'female'))
tico_7485$asthmahx <- factor(tico_7485$asthmahx, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$copdhx <- factor(tico_7485$copdhx, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$diabeteshx <- factor(tico_7485$diabeteshx, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$chfhx <- factor(tico_7485$chfhx, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$hypertenshx <- factor(tico_7485$hypertenshx, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$malignhx <- factor(tico_7485$malignhx, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$renalhx <- factor(tico_7485$renalhx, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$immuno_supp <- factor(tico_7485$immuno_supp, levels = c(0,1), labels = c('no', 'yes'))
tico_7485$ordpulmb1 <- factor(tico_7485$ordpulmb1, levels = c(2,3,4), labels = c('no 02', 'low 02', 'hi
tico_7485$corticobl <- factor(tico_7485$corticobl, levels = c(0,1), labels = c('no', 'yes'))

tico_7485 <- set_variable_labels(tico_7485, asthmahx = "Asthma",
                                copdhx = 'COPD',
                                diabeteshx = 'Diabetes requiring medication',
                                chfhx = 'CHF',
                                hypertenshx = 'Hypertension requiring medication',
                                malignhx = 'Malignancy (active or receiving treatment)',
                                renalhx = 'Renal impairment',
                                immuno_supp = 'Immunosuppressed',
                                newsbl = 'New Score',
                                creatbl = 'Creatinine (mg/dL)',
                                crpbl = 'C-reactive protein',
                                lymphbl = 'Lymphocytes (109/L)',
                                ordpulmb1 = 'Pulm ordinal',
                                vl_rslt_log10 = 'Log 10 of Viral load copies/mL')

# sapply(tico_7485, function(x) sum(is.na(x)))

mode_function <- function(x) {
  x <- as.character(x)
  tbl <- table(x, useNA = "ifany")
  as.character(names(tbl)[which.max(tbl)])
}

for (var in names(tico_7485)) {
  if (is.numeric(tico_7485[[var]])) {
    tico_7485[[var]][is.na(tico_7485[[var]])] <- median(tico_7485[[var]], na.rm = TRUE)
    print(paste("Imputed numeric variable:", var))
  } else if (is.factor(tico_7485[[var]])) {
    tico_7485[[var]][is.na(tico_7485[[var]])] <- mode_function(tico_7485[[var]])
    print(paste("Imputed categorical variable:", var))
  }
}

t1 <- CreateTableOne(var = names(tico_7485)[2:19], data = tico_7485, strata = 'corticobl')
t11 <- print(t1, smd = TRUE, showAllLevels = TRUE, varLabels = TRUE)
kable(t11, caption = "Summary Statistics Stratified by the use of corticosteroids")

```

Question 2

```
num_t1 <- table(tico_7485$corticobl)[2]
num_c1 <- table(tico_7485$corticobl)[1]

## for hospital_free_28
hf28_t <- tico_7485$hospital_free_28[tico_7485$corticobl == 'yes']
hf28_c <- tico_7485$hospital_free_28[tico_7485$corticobl == 'no']
et_hf28 <- mean(hf28_t, na.rm = T)
ec_hf28 <- mean(hf28_c, na.rm = T)

# Unadjusted ATE
ate_hf28 <- et_hf28 - ec_hf28
# Standard Error
se_hf28 <- sqrt(var(hf28_t, na.rm = T)/num_t1 + var(hf28_c, na.rm = T)/num_c1)
# Confidence Interval
conf_int_hf28 <- ate_hf28 + c(-1, 1)*qnorm(0.975)*se_hf28

output1 <- data.frame(
  Type = 'ATE',
  Estimate = round(ate_hf28, 3),
  SE = round(unname(se_hf28), 3),
  CI_Lower = round(conf_int_hf28[1], 3),
  CI_Upper = round(conf_int_hf28[2], 3)
)

kable(output1, caption = "Unadjusted ATE for Hospital-Free Days")
```

Question 3

```
calculate_ATE_ATT <- function(model_formula, data, num_bootstraps = 100) {

  # Fit initial outcome model
  hf28_model <- lm(model_formula, data = data)

  # Data for ATE (all individuals set to treated and control)
  data_trt <- data_con <- data
  data_trt$corticobl = 'yes'
  data_con$corticobl = 'no'

  # Predicted outcomes for ATE
  pred1_hf28 <- predict(hf28_model, newdata = data_trt, type = "response")
  pred0_hf28 <- predict(hf28_model, newdata = data_con, type = "response")
  ATE_hf28 <- mean(pred1_hf28 - pred0_hf28)

  # Data for ATT (only treated individuals set to control)
  data_att_trt <- data[data$corticobl == 'yes', ]
  data_att_con <- data_att_trt
  data_att_con$corticobl = 'no'

  # Predicted outcomes for ATT
```

```

pred1_att_hf28 <- predict(hf28_model, newdata = data_att_trt, type = "response")
pred0_att_hf28 <- predict(hf28_model, newdata = data_att_con, type = "response")
ATT_hf28 <- mean(pred1_att_hf28 - pred0_att_hf28)

# Bootstrap replication for SE
ate_hf28_boot <- c()
att_hf28_boot <- c()

for (i in 1:num_bootstraps) {
  data_boot <- data[sample(nrow(data), size = nrow(data), replace = TRUE), ]
  hf28_model_boot <- lm(model_formula, data = data_boot)

  # Bootstrap for ATE
  data_trt_boot <- data_con_boot <- data_boot
  data_trt_boot$corticobl = 'yes'
  data_con_boot$corticobl = 'no'
  pred1_hf28_boot <- predict(hf28_model_boot, newdata = data_trt_boot, type = "response")
  pred0_hf28_boot <- predict(hf28_model_boot, newdata = data_con_boot, type = "response")
  ate_hf28_boot <- c(ate_hf28_boot, mean(pred1_hf28_boot - pred0_hf28_boot))

  # Bootstrap for ATT
  data_att_boot_trt <- data_boot[data_boot$corticobl == 'yes', ]
  data_att_boot_con <- data_att_boot_trt
  data_att_boot_con$corticobl = 'no'
  pred1_att_hf28_boot <- predict(hf28_model_boot, newdata = data_att_boot_trt, type = "response")
  pred0_att_hf28_boot <- predict(hf28_model_boot, newdata = data_att_boot_con, type = "response")
  att_hf28_boot <- c(att_hf28_boot, mean(pred1_att_hf28_boot - pred0_att_hf28_boot))
}

# Calculate standard errors and confidence intervals
SE_ATE <- sd(ate_hf28_boot)
SE_ATT <- sd(att_hf28_boot)
CI_ATE_lower <- ATE_hf28 - qnorm(0.975) * SE_ATE
CI_ATE_upper <- ATE_hf28 + qnorm(0.975) * SE_ATE
CI_ATT_lower <- ATT_hf28 - qnorm(0.975) * SE_ATT
CI_ATT_upper <- ATT_hf28 + qnorm(0.975) * SE_ATT

# Return results as separate CI bounds
return(list(
  ATE = round(ATE_hf28, 3), SE_ATE = round(SE_ATE, 3),
  CI_ATE_lower = round(CI_ATE_lower, 3), CI_ATE_upper = round(CI_ATE_upper, 3),
  ATT = round(ATT_hf28, 3), SE_ATT = round(SE_ATT, 3),
  CI_ATT_lower = round(CI_ATT_lower, 3), CI_ATT_upper = round(CI_ATT_upper, 3)
))
}

full_model <- paste('hospital_free_28 ~ corticobl * (', paste(names(tico_7485)[1:19], collapse = ' + '),
full_hf28 <- lm(as.formula(full_model), data=tico_7485)

# do stepwise selection
stepwise_hf28 <- stepAIC(full_hf28, direction = "both")
# summary(stepwise_hf28)

```

```

select_hf28 <- hospital_free_28 ~ corticobl + gender + age + copdhx + malignhx + immuno_supp +
  bmi + newsbl + creatbl + crpbl + lymphbl + ordpulmb1 + vl_rslt_log10 + corticobl:bmi +
  corticobl:lymphbl
result_hf28 <- calculate_ATE_ATT(model_formula = select_hf28, data = tico_7485, num_bootstraps = 100)
# print(result_hf28)

output2 <- data.frame(
  Type = c("ATE", "ATT"),
  Estimate = c(result_hf28$ATE, result_hf28$ATT),
  SE = c(result_hf28$SE_ATE, result_hf28$SE_ATT),
  CI_Lower = c(result_hf28$CI_ATE_lower, result_hf28$CI_ATT_lower),
  CI_Upper = c(result_hf28$CI_ATE_upper, result_hf28$CI_ATT_upper)
)

kable(output2, caption = "ATE and ATT Estimates from Regression Model")

```

Question 4

```

# a) Find the IPW standardized mean difference among treatment groups
ps_model <- glm(corticobl ~ gender + age + copdhx + malignhx + immuno_supp +
  bmi + newsbl + creatbl + crpbl + lymphbl + ordpulmb1 + vl_rslt_log10, family = binomial(), data = tico_7485)
ps <- predict(ps_model, type = "response")
tico_7485$weight <- with(tico_7485, ifelse(corticobl == "yes", 1 / ps, 1 / (1 - ps)))

dataSvy <- svydesign(ids = ~ 1, data = tico_7485, weights = ~ weight)

tabWeighted <- svyCreateTableOne(vars = names(tico_7485)[2:19], strata = "corticobl", data = dataSvy,
  t2 <- print(tabWeighted, smd = TRUE, showAllLevels = TRUE, varLabels = TRUE)

kable(t2, caption = "Summary Statistics Stratified by the use of corticosteroids using IPW")

```

```

# b) ATE and ATT with PSS
# ATE estimate
tico_7485$corticobl_numeric <- as.numeric(tico_7485$corticobl == "yes")
ps <- predict(ps_model, type = "response")

ps_quintile_ate <- cut(ps, breaks = quantile(ps, probs = seq(0, 1, by = 0.2)),
  include.lowest = TRUE, labels = FALSE)
te_quintile <- tapply(tico_7485$hospital_free_28[tico_7485$corticobl == 'yes'],
  ps_quintile_ate[tico_7485$corticobl == 'yes'], mean) -
  tapply(tico_7485$hospital_free_28[tico_7485$corticobl == 'no'],
  ps_quintile_ate[tico_7485$corticobl == 'no'], mean)
ATE_PSS <- sum(te_quintile * table(ps_quintile_ate) / nrow(tico_7485), na.rm = TRUE)

# ATT estimate
ps_quintile_att <- cut(ps,
  breaks = c(0, quantile(ps[tico_7485$corticobl == 'yes'], p = c(0.2, 0.4, 0.6, 0.8)), 1), labels = 1
na <- nrow(tico_7485[tico_7485$corticobl == 'yes', ])
nAj <- table(ps_quintile_att[tico_7485$corticobl == 'yes'])
te_quintile_att <- tapply(tico_7485$hospital_free_28[tico_7485$corticobl == 'yes'],

```

```

        ps_quintile_att[tico_7485$corticobl == 'yes'], mean) -
    tapply(tico_7485$hospital_free_28[tico_7485$corticobl == 'no'],
        ps_quintile_att[tico_7485$corticobl == 'no'], mean)
ATT_PSS <- sum(te_quintile_att * nAj / nA, na.rm = TRUE)

# bootstrap to get the SE and CI
B <- 100
ATE_PSS.boot <- NULL
ATT_PSS.boot <- NULL
n <- nrow(tico_7485)

for(i in 1:B) {

    tico.boot <- tico_7485[sample(1:n, n, replace = TRUE), ]
    ps_model.boot <- glm(corticobl ~ gender + age + copdhx + malignhx + immuno_supp +
        bmi + newsbl + creatbl + crpbl + lymphbl + ordpulmb1 + vl_rslt_log10,
        data = tico.boot, family = "binomial")

    ps.boot <- predict(ps_model.boot, type = "response")

    # ATE
    ps_quintile.boot <- cut(ps.boot,
        breaks = quantile(ps.boot, probs = seq(0, 1, by = 0.2), na.rm = TRUE),
        include.lowest = TRUE, labels = FALSE)

    te_quintile.boot <- tapply(tico.boot$hospital_free_28[tico.boot$corticobl == 'yes'],
        ps_quintile.boot[tico.boot$corticobl == 'yes'], mean) -
        tapply(tico.boot$hospital_free_28[tico.boot$corticobl == 'no'],
            ps_quintile.boot[tico.boot$corticobl == 'no'], mean)

    nj.boot <- table(ps_quintile.boot)
    ATE.boot <- sum(te_quintile.boot * nj.boot / n, na.rm = TRUE)

    ATE_PSS.boot <- c(ATE_PSS.boot, ATE.boot)

    # ATT
    ps_quintile.boot_att <- cut(ps.boot,
        breaks = c(0, quantile(ps.boot[tico.boot$corticobl == 'yes'],
            probs = c(0.2, 0.4, 0.6, 0.8), na.rm = TRUE), 1),
        include.lowest = TRUE, labels = FALSE)

    nA.boot <- sum(tico.boot$corticobl == 'yes')
    nAj.boot <- table(ps_quintile.boot_att[tico.boot$corticobl == 'yes'])

    te_quintile_att.boot <- tapply(tico.boot$hospital_free_28[tico.boot$corticobl == 'yes'],
        ps_quintile.boot_att[tico.boot$corticobl == 'yes'], mean) -
        tapply(tico.boot$hospital_free_28[tico.boot$corticobl == 'no'],
            ps_quintile.boot_att[tico.boot$corticobl == 'no'], mean)

    ATT.boot <- sum(te_quintile_att.boot * nAj.boot / nA.boot, na.rm = TRUE)

    ATT_PSS.boot <- c(ATT_PSS.boot, ATT.boot)
}

```

```

SE_ATE_PSS <- sd(ATE_PSS.boot)
SE_ATT_PSS <- sd(ATT_PSS.boot)

CI_ATE_PSS <- ATE_PSS + c(-1, 1) * qnorm(0.975) * SE_ATE_PSS
CI_ATT_PSS <- ATT_PSS + c(-1, 1) * qnorm(0.975) * SE_ATT_PSS

output3 <- data.frame(
  Type = c("ATE", "ATT"),
  Estimate = c(ATE_PSS, ATT_PSS),
  SE = c(SE_ATE_PSS, SE_ATT_PSS),
  CI_Lower = c(CI_ATE_PSS[1], CI_ATT_PSS[1]),
  CI_Upper = c(CI_ATE_PSS[2], CI_ATT_PSS[2])
)

kable(output3, caption = "ATE and ATT Estimates using PSS")

```

```

# c) ATE and ATT estimates using IPW
# ATE
w1 <- as.numeric(tico_7485$corticobl_numeric)/ps
w0 <- (1-tico_7485$corticobl_numeric)/(1-ps)
ATE_IPW <- mean(tico_7485$hospital_free_28*w1) - mean(tico_7485$hospital_free_28*w0)

# ATT
w1 <- tico_7485$corticobl_numeric
w0 <- (1-tico_7485$corticobl_numeric)/(1-ps)*ps
ATT_IPW <- weighted.mean(tico_7485$hospital_free_28,w=w1) -
  weighted.mean(tico_7485$hospital_free_28,w=w0)

# bootstrap
B <- 100
n <- nrow(tico_7485)
ATE_IPW.boot <- NULL
ATT_IPW.boot <- NULL

for(i in 1:B) {

  tico.boot <- tico_7485[sample(1:n, n, replace = TRUE), ]

  p1.boot <- glm(corticobl ~ gender + age + copd hx + malign hx + immuno_supp + bmi + newsbl +
    creatbl + crpbl + lymphbl + ordpulmbl + vl_rslt_log10, data = tico.boot, family = "b
  ps.boot <- predict(p1.boot, type = "response")

  w1.boot <- as.numeric(tico.boot$corticobl_numeric) / ps.boot
  w0.boot <- (1 - tico.boot$corticobl_numeric) / (1 - ps.boot)

  ATE_IPW.boot <- c(ATE_IPW.boot,
    mean(tico.boot$hospital_free_28 * w1.boot) -
    mean(tico.boot$hospital_free_28 * w0.boot))

  w1_att.boot <- tico.boot$corticobl_numeric
  w0_att.boot <- (1 - tico.boot$corticobl_numeric) / (1 - ps.boot) * ps.boot

  ATT_IPW.boot <- c(ATT_IPW.boot,

```

```

        weighted.mean(tico.boot$hospital_free_28, w = w1_att.boot) -
        weighted.mean(tico.boot$hospital_free_28, w = w0_att.boot))
}

SE_ATE_IPW <- sd(ATE_IPW.boot)
CI_ATE_IPW <- ATE_IPW + c(-1, 1) * qnorm(0.975) * SE_ATE_IPW

SE_ATT_IPW <- sd(ATT_IPW.boot)
CI_ATT_IPW <- ATT_IPW + c(-1, 1) * qnorm(0.975) * SE_ATT_IPW

output4 <- data.frame(
  Type = c("ATE", "ATT"),
  Estimate = c(ATE_IPW, ATT_IPW),
  SE = c(SE_ATE_IPW, SE_ATT_IPW),
  CI_Lower = c(CI_ATE_IPW[1], CI_ATT_IPW[1]),
  CI_Upper = c(CI_ATE_IPW[2], CI_ATT_IPW[2])
)

kable(output4, caption = "ATE and ATT Estimates using IPW")

```

```

# d) 1:1 matching
# i. Find the standardized mean difference among the matched cohort.
rownames(tico_7485) <- 1:nrow(tico_7485)
mod_match <- matchit(corticobl ~ gender + age + copdhx + malignhx + immuno_supp + bmi + newsbl + creatbl +
  ordpulmbl + vl_rslt_log10, distance="logit", method="nearest", data=tico_7485, ratio=1)
#print(tail(mod_match$match.matrix))

matched_pairs <- na.omit(mod_match$match.matrix) # to check if controls re used or not
n_matched <- nrow(matched_pairs)
print(n_matched)

tico_match <- data.frame(rbind(tico_7485[tico_7485$corticobl == 'yes', ],
  tico_7485[as.numeric(mod_match$match.matrix), ]))

tico_match <- set_variable_labels(tico_match, asthmahx = "Asthma",
  copdhx = 'COPD',
  diabeteshx = 'Diabetes requiring medication',
  chfhx = 'CHF',
  hypertenshx = 'Hypertension requiring medication',
  malignhx = 'Malignancy (active or receiving treatment)',
  renalhx = 'Renal impairment',
  immuno_supp = 'Immunosuppressed',
  newsbl = 'New Score',
  creatbl = 'Creatinine (mg/dL)',
  crpbl = 'C-reactive protein',
  lymphbl = 'Lymphocytes (10^9/L)',
  ordpulmbl = 'Pulm ordinal',
  vl_rslt_log10 = 'Log 10 of Viral load copies/mL')

tabmatched1.1 <- CreateTableOne(vars = names(tico_7485)[2:19], strata = "corticobl",
  data = tico_match, test = FALSE)
t3 <- print(tabmatched1.1, smd = TRUE, showAllLevels = TRUE,
  varLabels = TRUE)

```

```

kable(t3, caption = "Summary Statistics Stratified by the use of corticosteroids using 1:1 Matching")

# ii.Estimate the ATT using 1:1 matching
p1 <- mean(tico_7485$hospital_free_28[tico_7485$corticobl == 'yes'])
p0 <- mean(tico_7485$hospital_free_28[as.numeric(mod_match$match.matrix)],na.rm=TRUE)
n1 <- table(tico_7485$corticobl)[2]
n0 <- length(as.numeric(mod_match$match.matrix))

ATTmatch <- p1 - p0
SE1match <- sqrt(-1*(p1*(1-p1)/n1 + p0*(1-p0)/n0))
CI1match <- p1 - p0 + c(-1, 1)*qnorm(0.975)*SE1match

output5 <- data.frame(
  Type = 'ATT',
  Estimate = round(ATTmatch, 3),
  SE = round(unname(SE1match), 3),
  CI_Lower = round(CI1match[1], 3),
  CI_Upper = round(CI1match[2], 3)
)

kable(output5, caption = "ATT Estimate using 1:1 Matching")

```